

Research Notes on Curse of dimensionality

Curse of dimensionality

$$V_d(r) = \frac{\pi^{d/2}}{\Gamma(d/2+1)} r^d$$

>> Section 1

Combinatorics In some problems, each variable can take one of several discrete values, or the range of possible values is divided to give a finite number of possibilities. Taking the variables together, a huge number of combinations of values must be considered. This effect is also known as the combinatorial explosion. Even in the simplest case of d binary variables, the number of possible combinations already is 2^d , exponential in the dimensionality. Naively, each additional dimension doubles the effort needed to try all combinations.

>> Section 2

V hypersphere V hypercube $= \pi^{d/2} / 2^d d^{-1} \Gamma(d/2) \rightarrow 0$ as $d \rightarrow \infty$. Furthermore, the distance between the center and the corners is $r \sqrt{d}$, which increases without bound for fixed r . In this sense when points are uniformly generated in a high-dimensional hypercube, almost all points are much farther than r units away from the center. In high dimensions, the volume of the d -dimensional unit hypercube (with coordinates of the vertices ± 1) is concentrated near a sphere with the radius $d/3$ for large dimension d . Indeed, for each coordinate x_i the average value of x_i^2 in the cube is

>> Section 3

Concentration of scores and distances: derived values such as distances become numerically similar
Irrelevant attributes: in high dimensional data, a significant number of attributes may be irrelevant
Definition of reference sets: for local methods, reference sets are often nearest-neighbor based
Incomparable scores for different dimensionalities: different subspaces produce incomparable scores
Interpretability of scores: the scores often no longer convey a semantic meaning
Exponential search space: the search space can no longer be systematically scanned
Data snooping bias: given the large search space, for every desired significance a hypothesis can be found
Hubness: certain objects occur more frequently in neighbor lists than others. Many of the analyzed specialized methods tackle

one or another of these problems, but there remain many open research questions.

>> Section 4

The curse of dimensionality refers to various phenomena that arise when analyzing and organizing data in high-dimensional spaces that do not occur in low-dimensional settings such as the three-dimensional physical space of everyday experience. The expression was coined by Richard E. Bellman when considering problems in dynamic programming. The curse generally refers to issues that arise when the number of datapoints is small (in a suitably defined sense) relative to the intrinsic dimension of the data. Dimensionally cursed phenomena occur in domains such as numerical analysis, sampling, combinatorics, machine learning, data mining and databases. The common theme of these problems is that when the dimensionality increases, the volume of the space increases so fast that the available data become sparse. In order to obtain a reliable result, the amount of data needed often grows exponentially with the dimensionality. Also, organizing and searching data often relies on detecting areas where objects form groups with similar properties; in high dimensional data, however, all objects appear to be sparse and dissimilar in many ways, which prevents common data organization strategies from being efficient.

>> Section 5

Blessing of dimensionality Despite the expected "curse of dimensionality" difficulties, common-sense heuristics based on the most straightforward methods "can yield results which are almost surely optimal" for high-dimensional problems. The term "blessing of dimensionality" was introduced in the late 1990s. Donoho in his "Millennium manifesto" explained why he thinks the "blessing of dimensionality" will form a basis of future data mining. The effects of the blessing of dimensionality were discovered in many applications and found their foundation in the concentration of measure phenomena. One example of the blessing of dimensionality phenomenon is linear separability of a random point from a large finite random set with high probability even if this set is exponentially large: the number of elements in this random set can grow exponentially with dimension. Moreover, this linear functional can be selected in the form of the simplest linear Fisher discriminant. This separability theorem was proven for a wide class of probability distributions: general uniformly log-concave distributions, product distributions in a cube and many other families (reviewed recently in). "The blessing of dimensionality and the curse of dimensionality are two sides of the same coin." For example, the typical property of essentially

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high-dimensional probability distributions in a high-dimensional space is: the squared distance of random points to a selected point is, with high probability, close to the average (or median) squared distance. This property significantly simplifies the expected geometry of data and indexing of high-dimensional data (blessing), but, at the same time, it makes the similarity search in high dimensions difficult and even useless (curse). Zimek et al. noted that while the typical formalizations of the curse of dimensionality affect i.i.d. data, having data that is separated in each attribute becomes easier even in high dimensions, and argued that the signal-to-noise ratio matters: data becomes easier with each attribute that adds signal, and harder with attributes that only add noise (irrelevant error) to the data. In particular for unsupervised data analysis this effect is known as swamping.

>> Section 6

Sampling There is an exponential increase in volume associated with adding extra dimensions to a mathematical space. For example, $10^2 = 100$ evenly spaced sample points suffice to sample a unit interval (try to visualize a "1-dimensional" cube, i.e. a line) with no more than $10^{-2} = 0.01$ distance between points; an equivalent sampling of a 10-dimensional unit hypercube with a lattice that has a spacing of $10^{-2} = 0.01$ between adjacent points would require $10^{20} = [(10^2)^{10}]$ sample points. In general, with a spacing distance of 10^{-n} the 10-dimensional hypercube appears to be a factor of $10^n(10-1) = [(10^n)10/(10^n)]$ "larger" than the 1-dimensional hypercube, which is the unit interval. In the above example $n = 2$: when using a sampling distance of 0.01 the 10-dimensional hypercube appears to be 10^{18} "larger" than the unit interval. This effect is a combination of the combinatorics problems above and the distance function problems explained below.